ARTHENTIC PROTOCOL | MASTER ARCHITECTURE & IMPLEMENTATION BLUEPRINT

Next-Steps Roadmap, 6-Layer System Blueprint & Unified Monorepo Architecture (Mobile + Web + Chain)

Synthesized from UML Part 1 (11 Modules, 9 Actors), Sequence & DB Design (SD-01..12, 40 PG Tables, 6 Redis Keys), Locked ADRs (ADR-01..09), and Current Prototype Audit (Flutter iOS/Web + FastAPI Modular Monolith + Base L2 / Eth L1)

Document Scope: Exhaustive engineering and architectural blueprint derived from ARTHENTIC_MASTER_IMPLEMENTATION_AND_BUSINESS_PLAN.md, Arthentic_UML_Diagrams_Part_1.pdf (9 Actors, 11 Use-Case Modules UC-A-UC-L, 3 Class Diagrams, 48 Activity Diagrams), Arthentic_Sequence_and_Database_Design.pdf (12 Sequence Diagrams SD-01-SD-12, 40 PostgreSQL 16 Tables across 6 ERDs, 6 Redis Hot-Path Keys, 9 Core Invariants), and the current Arthenthic-Mobile repository (app/ Flutter + server/ FastAPI).

Core Questions Answered:
1. What is the current implementation state and what are the exact next steps?
2. What other infrastructure, cryptographic, and blockchain layers need to be built?
3. How does the Web Client fit in, and how can Mobile, Web, Backend, Protocol Node, Indexer, Paymaster, and Smart Contracts all be implemented in one unified codebase?

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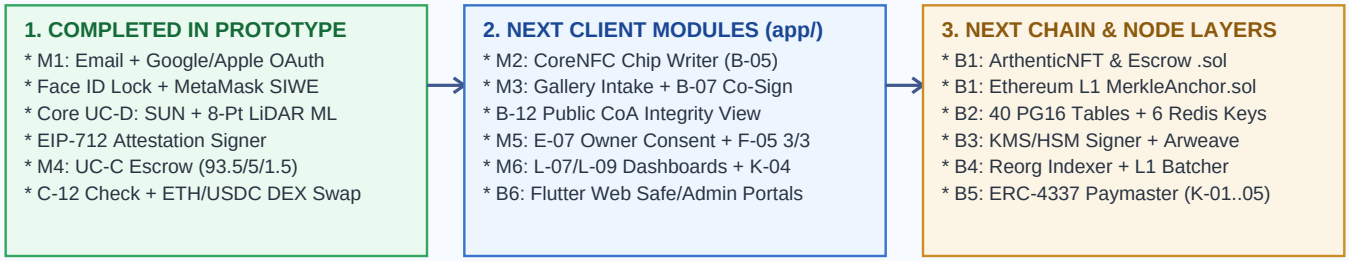
* Complete Unified Monorepo Directory Structure
* 8-Week Execution & Integration Schedule

1. Executive Summary & Current Codebase Audit Matrix

1.1 Current State of Arthenthic-Mobile

The current repository already implements a strong vertical slice of the Mobile App (app/) and FastAPI Backend (server/), backed by 65 passing backend unit/integration tests and 9 locked Architectural Decision Records (ADR-01 through ADR-09).

ARCHITECTURE DIAGRAM 1: PROTOTYPE STATUS VS. NEXT DELIVERY TRACKS



1.2 Detailed Feature & File Audit Matrix

Use-Case Module & Spec IDs	Existing Files in Arthenthic-Mobile	Completion %	Exact Remaining Engineering Work
UC-A: Accounts, Auth, OAuth & Wallets A-01, A-05-A-10 | SD-01 | ERD 1 (7 tables)	server/routes/auth.py server/auth/metamask.py server/routes/roles.py app/lib/features/auth/* app/lib/features/wallet/*	85%	- Wire live WalletConnect v2 (reown_appkit) deep-link callbacks on physical iOS. - Add ERC-4337 Passkey Smart Wallet factory (Coinbase Smart Wallet). - Sync on-chain RoleGranted/RoleRevoked events into role_grants.
UC-B: Artwork Intake & Genesis Mint B-01-B-05, B-07, B-08, B-12 | SD-02 | ERD 2 (10 tables)	server/routes/genesis.py app/lib/features/scan/scan_screen.dart	50%	- Separate Gallery Intake Dossier (B-01) from Artist EIP-712 Co-Sign (B-07) and Gallery Safe genesisMint() (B-08). - Add CoreNFC NTAG 424 DNA key provisioning (B-05) and Arweave/IPFS upload (B-04). - Build Certificate of Authenticity Screen (B-12) verifying keccak256(metadata) == genesisHash.
UC-C: Escrow Trading, Tap Claim & DEX Swap C-01, C-03*, C-04, C-06, C-10, C-12 | SD-03, SD-04 | ERD 3 (3 tables)	server/routes/marketplace.py app/lib/features/marketplace/marketplace_screen.dart server/tests/test_marketplace.py	90%	- Connect off-chain marketplace state to deployed ArthenticEscrow.sol on Base L2 Sepolia. - Add shipments carrier tracking webhook/UI and automated post-expiry refund() trigger.
UC-D: Authenticity Verification (L1 & L2) D-01, D-02, D-04-D-06, D-09, D-14*, D-15* | SD-05, SD-06 | ERD 4	server/routes/verify.py server/crypto/sun.py server/ml/embedder.py app/lib/features/result/result_screen.dart	80%	- Move SUN counter lock to Redis (sun:ctr:{chipId}) + atomic SQL check. - Build StatusBadgeWidget (D-15*: Clear, Destroyed, Flagged, Chip-Replaced, Unsynchronized Ownership). - Wire on-chain recordVerification() transaction (D-04).
UC-E: Conservation & Condition Updates E-01-E-04, E-06, E-07* | SD-07 | ERD 5 (4 tables)	server/routes/genesis.py (/v1/conservation/* stubs)	35%	- Build Owner-to-Conservator request workflow (E-01). - Build Collector EIP-712 Owner Consent Screen (E-07* / ADR-03). - Build side-by-side Genesis vs. Restoration History View (E-06).

Use-Case Module & Spec IDs	Existing Files in Arthenthic-Mobile	Completion %	Exact Remaining Engineering Work
UC-F: Damaged NFC Chip Replacement F-05 | SD-08 | ERD 2 (chip_replacements, chip_approvals)	server/auth/metamask.py (F-05 action definition)	25%	- Build 3-of-3 Multi-Approval Tracker UI & API (F-05) collecting Collector (1/3), Artist (2/3), and Verifier (3/3) EIP-712 signatures + L2 optical proof (>= 8800 bps).
UC-H & UC-I: Provenance, Attestation & L1 Anchor H-03, H-05, I-01, I-02 | SD-10, SD-12 | ERD 4	server/crypto/attestation.py app/lib/features/provenance/provenance_screen.dart	55%	- Deploy MerkleAnchor.sol on Ethereum L1 Sepolia (H-05). Build background batch worker (every 1,000 events or 24h) and Merkle inclusion proof badge on ProvenanceScreen.
UC-J, UC-K & UC-L: Governance, Paymaster & Dashboards J-01, J-05, K-01-K-05, L-07, L-09 | SD-09, SD-11 | ERD 6	server/routes/admin.py app/lib/features/home/home_screen.dart	40%	- Build ERC-4337 Paymaster policy/quota engine (K-01-K-05) & Self-Pay fallback modal (K-04). - Build dedicated Artist (L-07) & Collector (L-09) dashboards. - Build Admin Incident Console (J-05) & Gallery Safe Web Console.

2. Part I: Prioritized Next Steps -- Step-by-Step Engineering Roadmap

To transition from the current prototype to a full end-to-end Base L2 + Ethereum L1 testnet deployment, development should proceed in four sequential engineering steps.

2.1 Step 1: Smart Contract Layer (Phase B1 -- Base L2 & Ethereum L1)

The highest-leverage next step is creating the contracts/ workspace using Foundry (forge) and Solidity ^0.8.24 with OpenZeppelin v5 Upgradeable contracts. All 9 locked ADRs (ADR-01-ADR-09) directly shape these contracts:

1. contracts/src/Roles.sol

Defines role constants shared across ArthenticNFT and ArthenticEscrow:

CODE / STRUCTURE (SOLIDITY)

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.24;

library Roles {
    bytes32 internal constant ARTIST_ROLE        = keccak256("ARTIST_ROLE");
    bytes32 internal constant VERIFIER_ROLE      = keccak256("VERIFIER_ROLE");
    bytes32 internal constant CONSERVATOR_ROLE   = keccak256("CONSERVATOR_ROLE");
    bytes32 internal constant PROTOCOL_NODE_ROLE = keccak256("PROTOCOL_NODE_ROLE");
    bytes32 internal constant ANCHORER_ROLE      = keccak256("ANCHORER_ROLE");
}
```

2. contracts/src/AttestationVerifier.sol

Abstract EIP-712 verifier inherited by both ArthenticNFT and ArthenticEscrow:

* **Constants:** uint16 public constant MIN_SIMILARITY_BPS = 8800; (88.0% optical + LiDAR surface match threshold).
* **State:** mapping(uint256 => bool) public usedNonce;
* **Struct:**

CODE / STRUCTURE (SOLIDITY)

```
  struct Attestation {
      uint256 tokenId;
      bytes32 sunDigest;      // keccak256(abi.encodePacked(uid, counter, cmac))
      uint16  similarityBps;  // 0 for L1 NFC tap; 8800..10000 for L2 Optical scan
      uint64  deadline;       // block.timestamp + 300s short validity window
      uint256 nonce;          // single-use anti-replay random uint256
  }
```

* **Verification Invariants (_verifyAttestation):**
  1. Reverts with ATTESTATION_EXPIRED() if block.timestamp > a.deadline.
  2. Reverts with REPLAY() if usedNonce[a.nonce] is true.
  3. Reverts with SIMILARITY_BELOW_THRESHOLD() if requireOptical && a.similarityBps < MIN_SIMILARITY_BPS.
  4. Recovers signer = ECDSA.recover(_hashTypedDataV4(structHash), sig) and reverts with INVALID_SIGNER() unless hasRole(Roles.PROTOCOL_NODE_ROLE, signer).
  5. Marks usedNonce[a.nonce] = true before any external call (Checks-Effects-Interactions).

3. contracts/src/ArthenticNFT.sol (Base L2 -- UUPS Upgradeable)

* **Inheritance:** UUPSUpgradeable, ERC721URIStorageUpgradeable, ERC2981Upgradeable, AccessControlUpgradeable, PausableUpgradeable, AttestationVerifier.
* **Core State:**
  * mapping(bytes32 => bool) public chipBound; (Invariant SC-CHIP-01: a physical chip can never be bound to two artworks).
  * mapping(uint256 => bytes32) public genesisHash; (Immutable once set in genesisMint -- Invariant SC-GEN-01).
  * mapping(uint256 => bytes32) public activeChip;
  * mapping(uint256 => bool) public destroyed;
* **Key Functions & ADR Enforcement:**
  * genesisMint(address artist, bytes32 genesisVectorHash, bytes32 chipIdHash, string calldata metadataURI, bytes calldata artistSig) (onlyRole(VERIFIER_ROLE)):
    * Enforces hasRole(ARTIST_ROLE, artist), !chipBound[chipIdHash], and genesisVectorHash != bytes32(0).
    * **ADR-02 Enforcement:**Verifies ECDSA.recover(genesisDossierDigest, artistSig) == artist on-chain so a Gallery cannot mint without the Artist's EIP-712 co-signature (B-07).
    * Mints tokenId to artist, sets _setTokenRoyalty(tokenId, artist, 500) (5.0% ERC-2981 royalty), and emits GenesisRegistered.
  * updateCondition(uint256 tokenId, bytes32 newVectorHash, string calldata noteURI, bytes calldata ownerConsentSig) (onlyRole(CONSERVATOR_ROLE)):
    * Enforces !destroyed[tokenId] and leaves genesisHash[tokenId] untouched.
    * **ADR-03 Enforcement:**Verifies ECDSA.recover(consentDigest, ownerConsentSig) == ownerOf(tokenId) on-chain (E-07*).
    * Emits ConditionUpdated(tokenId, newVectorHash, msg.sender, noteURI, block.timestamp).
  * replaceChip(uint256 tokenId, bytes32 newChipIdHash, Attestation calldata optical, bytes calldata nodeSig, bytes[3] calldata approvals) (onlyRole(VERIFIER_ROLE)):
    * Calls _verifyAttestation(optical, nodeSig, true) (similarityBps >= 8800), checks !chipBound[newChipIdHash], and verifies 3 distinct EIP-712 signatures from ownerOf(tokenId) (Collector), VERIFIER_ROLE (Gallery), and original ARTIST_ROLE (F-05).
    * Unbinds old chip, binds newChipIdHash, and emits ChipReplaced.

4. contracts/src/ArthenticEscrow.sol (Base L2 -- Pausable, ReentrancyGuard)

* **Key Functions & ADR Enforcement:**
  * createListing(uint256 tokenId, address currency, uint256 price, uint64 expiry): Transfers NFT into escrow custody (status = Status.Listed). Supports native Base ETH (address(0)) and USDC (ADR-04).
  * **cancelListing(uint256 listingId):** Seller reclaims NFT if status == Status.Listed (C-03*).
  * deposit(uint256 listingId, bytes calldata permit2Data) (payable): Buyer locks funds (status = Status.Funded, C-04).
  * claim(uint256 listingId, Attestation calldata a, bytes calldata sig) (nonReentrant, C-06):
    * Verifies msg.sender == listing.buyer, status == Status.Funded, and _verifyAttestation(a, sig, false).
    * **ADR-05 3-Way Atomic Split:** Calculates artistRoyalty = (price * 500) / 10000 (5.0%), protocolFee = (price * 150) / 10000 (1.5% to treasurySafe), and sellerNet = price - artistRoyalty - protocolFee (93.5% to seller).

- Transfers NFT to buyer, distributes funds, and emits ClaimCompleted.
- **refund(uint256 listingId) (nonReentrant):** If block.timestamp > listing.expiry and status == Status.Funded, returns 100% of funds to buyer and returns the NFT to seller.

5. contracts/src/MerkleAnchor.sol (Ethereum L1)

- Anchors batched Base L2 provenance events onto Ethereum L1 (H-05):

CODE / STRUCTURE (SOLIDITY)

```
function anchorRoot(bytes32 root, uint64 fromBlock, uint64 toBlock)
    external onlyRole(Roles.ANCHORER_ROLE)
{
    emit RootAnchored(root, fromBlock, toBlock, uint64(block.timestamp));
}
```

2.2 Step 2: Hardware Sensor Pipeline, Multi-Actor Genesis Intake & CoA (Phases M2 & M3 / UC-B)

Currently, POST /v1/genesis/prepare in server/routes/genesis.py combines intake and preparation into a single step. To match SD-02 and UC-B:

1. CoreNFC NTAG 424 DNA Provisioning & Parsing (app/lib/services/nfc_service.dart):
   - **Reader Mode (D-01):** Parses dynamic NDEF URL https://verify.arthentic.io/?uid=<7B_hex>&ctr=<3B_hex>&cmac=<8B_hex> generated on every tap.
   - **Provisioning Writer Mode (B-05):** Used by Galleries during intake to read the tag's 7-byte UID, compute chipIdHash = keccak256(uid), query ArthenticNFT.chipBound(chipIdHash) to prevent duplicate binding, and configure SDM (Secure Dynamic Messaging) keys via KMS.
2. Three-Stage Genesis UI & API (app/lib/features/genesis/ & server/routes/genesis.py):
   - **Stage 1 -- Gallery Intake Dossier (B-01-B-05):** Gallery enters artwork metadata (title, artist_wallet, medium, width_cm, height_cm, year_created, condition_notes), captures 5-10 micro-point macro + LiDAR depth scans (B-02), uploads immutable metadata/vectors to Arweave/IPFS (B-04), and binds the NFC chip (B-05). Status becomes AWAITING_ARTIST_COSIGN.
   - **Stage 2 -- Artist EIP-712 Co-Signature (B-07):** Artist sees a pending co-sign card on their Home/Wallet screen, inspects the human-readable dossier (genesisVectorHash, chipIdHash, 5.0% ERC-2981 royalty), and signs via MetaMask (eth_signTypedData_v4). Status becomes READY_FOR_SAFE_MINT.
   - Stage 3 -- Gallery Safe Mint (B-08) & Public CoA Screen (B-12): Gallery executes genesisMint(). Any user opening the Certificate of Authenticity Screen (B-12) sees the full provenance dossier while the client recomputes keccak256(canonicalMetadata) against onchain.genesisHash(tokenId) and renders a green "Cryptographic Hash Verified" badge (or a red "Metadata Integrity Mismatch" warning if altered).

2.3 Step 3: Post-Sale Restoration Lifecycle & 3-of-3 Chip Swap (Phase M5 / UC-E & UC-F)

1. End-to-End Restoration Workflow (E-01-E-07*, SD-07):
   - **POST /v1/conservation/requests (E-01):** Collector selects an accredited Conservator (hasRole(CONSERVATOR_ROLE)) and submits damage photos/notes.
   - POST /v1/conservation/requests/{id}/scan (E-02, E-04): Conservator completes physical restoration, re-scans the 5-10 micro-points (newVectorHash), and uploads the treatment report (noteURI) to Arweave.
   - POST /v1/conservation/requests/{id}/owner-consent (E-07* / ADR-03): Collector reviews the before/after visual comparison and signs an EIP-712 OwnerConsent(tokenId, newVectorHash, noteURI, nonce, deadline) payload in MetaMask.
   - POST /v1/conservation/requests/{id}/finalize (E-03): Conservator submits ArthenticNFT.updateCondition(tokenId, newVectorHash, noteURI, ownerConsentSig).
   - **Restoration Timeline UI (E-06):** Displays the immutable genesisHash alongside every chronological ConditionUpdated entry so future L2 scans compare against the latest authorized post-restoration vector while preserving original creation proof.
2. **3-of-3 Chip Replacement Workflow (F-05, SD-08):**

- When a physical NTAG 424 DNA tag is damaged, no single actor can swap the chip (preventing a rogue gallery or owner from moving an NFT identity onto a fake painting):
- **Optical Gate:** Gallery performs an L2 8-point LiDAR/macro scan of the canvas; Protocol Node verifies similarityBps >= 8800 against genesisHash and issues an optical Attestation.
- 3-of-3 Approval Tracker (chip_replacements + chip_approvals tables):
  - **Approval 1/3:** Current Collector (ownerOf(tokenId)) signs EIP-712 ChipReplacementApproval.
  - **Approval 2/3:** Original Artist (ARTIST_ROLE) signs EIP-712 ChipReplacementApproval.
  - **Approval 3/3:** Accredited Verifier Gallery (VERIFIER_ROLE) signs EIP-712 ChipReplacementApproval and executes replaceChip(tokenId, newChipIdHash, opticalAttestation, nodeSig, approvals).

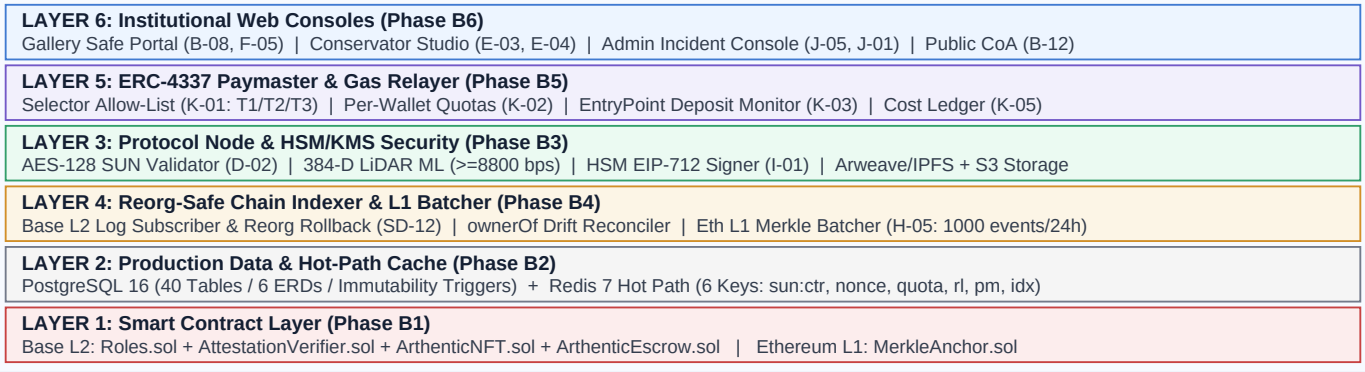
2.4 Step 4: Role Dashboards, Status Badges & Paymaster Fallback (Phase M6 / UC-D, UC-K, UC-L)

1. **Unified Artwork Status Badge System (D-15*):**
   - Every scan result, CoA, and marketplace card renders deterministic status badges:
     - **CLEAR (Green):** Active chip bound, ownerOf synchronized, L1/L2 verification passed.
     - **CHIP_REPLACED (Blue Info):** Chip replaced via F-05 3-of-3 governance; full audit trail linked.
     - **UNSYNCHRONIZED_OWNERSHIP (C-12 Amber/Red):** Physical possessor or seller wallet does not match on-chain ArthenticNFT.ownerOf(tokenId) -- blocks off-chain payment.
     - **FLAGGED (Orange):** Under active J-05 security investigation or anomalous SUN replay burst.
     - **DESTROYED (Red):** Marked destroyed on-chain via markDestroyed(tokenId, proofURI) -- permanently blocks listing, deposit, and condition updates.
2. Artist Dashboard (L-07) & Collector Dashboard (L-09):
   - **Artist View (L-07):** Total works minted, active primary/secondary listings, and historical + claimable 5% ERC-2981 royalties indexed from ClaimCompleted events.
   - **Collector View (L-09):** Portfolio valuation, active escrow deliveries with countdown timers to expiry (enabling 1-click refund() if undelivered), restoration history, and verification log.
3. **Paymaster Self-Pay Gas Fallback Modal (K-04):**
   - When the ERC-4337 Paymaster declines sponsorship (quota K-02 exhausted, non-sponsored tier, or low Paymaster deposit K-03), the client intercepts the error code, estimates Base L2 gas (~0.000005 ETH / ~$0.01), and presents a 1-tap "Pay Gas Directly via MetaMask / Smart Wallet" confirmation sheet.

3. Part II: Deep Dive into the 6 Additional Architecture Layers (Phases B1-B6)

Beyond the mobile UI (app/) and BFF routes (server/routes/), the complete Arthentic specification defines 6 specialized infrastructure layers.

ARCHITECTURE DIAGRAM 3: THE 6 PROTOCOL & INFRASTRUCTURE LAYERS (PHASES B1-B6)



3.1 Layer 1: Smart Contract Layer (contracts/ -- Phase B1)

* **Runtime:** Base L2 (Chain ID 8453 Mainnet / 84532 Base Sepolia) for low-latency, sub-cent transactions (ArthenticNFT, ArthenticEscrow) + Ethereum L1 (Chain ID 1 Mainnet / 11155111 Sepolia) for ultimate finality (MerkleAnchor).
* **Upgradeability & Security:** ArthenticNFT uses OpenZeppelin UUPSUpgradeable controlled by the Protocol Admin Timelock + Gnosis Safe (DEFAULT_ADMIN_ROLE), while genesisHash[tokenId] is protected by a write-once check (if (genesisHash[tokenId] != bytes32(0)) revert GENESIS_IMMUTABLE()).

3.2 Layer 2: Production Data & Hot-Path Cache Layer (PostgreSQL 16 + Redis 7 -- Phase B2)

The specification (Arthentic_Sequence_and_Database_Design.pdf, Part 2) defines 40 PostgreSQL 16 tables across 6 domains and 6 Redis hot-path keys:

The 6 PostgreSQL 16 Domains (40 Tables)

1. **Domain 1 -- Identity & Access (7 tables):**

users, wallets (wallet_type IN ('smart_wallet', 'safe', 'eoa')), institutions, roles, role_grants, artist_applications, notifications.

2. **Domain 2 -- Artwork, Genesis & Chips (10 tables):**

stored_files, artworks (protected by forbid_genesis_change() trigger), intake_dossiers, artwork_files, scan_sessions, scan_points, feature_vectors, nfc_chips (with one_active_chip partial unique index and KMS-encrypted uid_enc), chip_replacements, chip_approvals (UNIQUE (replacement_id, party)).

3. **Domain 3 -- Marketplace & Escrow (3 tables):**

listings (protected by state-transition trigger enforcing Listed -> Funded -> Completed | Refunded | Cancelled), payments, shipments.

4. Domain 4 -- Verification, Attestation & Provenance (8 tables):

verification_requests (monthly range-partitioned), sun_validations, attestations (CHECK (purpose <> 'l2' OR similarity_bps >= 8800)), verification_records, chain_events (partitioned), provenance_events, merkle_batches, indexer_cursors.

5. **Domain 5 -- Restoration (4 tables):**

restoration_requests, restoration_reports, owner_consents, condition_updates.

6. Domain 6 -- Governance, Paymaster & Audit (8 tables):

signer_keys, incidents, incident_actions, audit_log (REVOKE UPDATE, DELETE append-only), paymaster_policies, user_quotas, sponsored_txs, paymaster_balance_snapshots.

The 6 Redis 7 Hot-Path Keys

Redis Key Pattern	Data Type & TTL	Atomic Operation & Security Purpose		
1. sun:ctr:{chipId}	STRING (int, no expiry)	Monotonic Lua check-and-set (new_ctr > old_ctr) rejecting replayed NFC SUN URLs in < 1 ms before hitting PostgreSQL (D-02).		
2. nonce:{nonce}	STRING (TTL = deadline - now + 60s)	SET nonce:{n} 1 NX EX <ttl> at EIP-712 signing time (I-01) so two concurrent requests never receive the same attestation nonce.		
3. quota:{wallet}:{poli-cyId}:{period}	STRING (TTL = period_end)	INCRBY atomic counter enforcing per-user Paymaster gas sponsorship limits (K-02).		

Redis Key Pattern	Data Type & TTL	Atomic Operation & Security Purpose		
4. rl:verify:{ipHash}	ZSET / STRING (TTL = 60s)	Sliding-window rate limiter protecting public wallet-less verification (D-14*) from scraping or DoS.		
5. pm:balance	HASH (TTL = 30s)	Cached EntryPoint deposit status (status: ok	warning	empty, balance_wei) consulted before signing any paymasterAndData (K-03).
6. idx:head:{chainId}	STRING (no expiry)	Latest indexed block number and timestamp for health checks and lag alerts (SD-12).		

3.3 Layer 3: Protocol Node & HSM/KMS Security Layer (server/node/ -- Phase B3)

* Zero Plaintext Keys in DB or RAM (SC-NODE-01, SC-NODE-02):
  * nfc_chips.key_ref stores only an opaque KMS/HSM key handle (arn:aws:kms:... or Vault Transit path). AES-128 CMAC verification happens inside the KMS/HSM boundary.
  * signer_keys.hsm_key_ref references the secp256k1 key holding PROTOCOL_NODE_ROLE. When the Protocol Node signs an EIP-712 Attestation, it sends only the 32-byte digest to HSM/KMS and receives (r, s, v).
* Permanent Decentralized Storage Adapter (server/node/storage.py):
  * Uploads canonical JSON metadata, 384-D feature vectors, CoA documents, and conservation reports to Arweave / IPFS (B-04, E-04), returning ar://... or ipfs://... URIs and verifying keccak256 content hashes before on-chain calls.

3.4 Layer 4: Reorg-Safe Chain Event Indexer & L1 Merkle Batcher (server/indexer/ -- Phase B4)

1. **Reorg-Safe Event Indexing (SD-12):**
   * Polls or subscribes via WebSocket to ArthenticNFT and ArthenticEscrow on Base L2.
   * Inserts raw logs into chain_events with UNIQUE (chain_id, tx_hash, log_index) and finalized = false.
   * Checks parent block_hash continuity against indexer_cursors.last_block_hash. If an L2 reorg occurs, deletes unfinalized rows (block_number >= fork_block AND finalized = false), rewinds indexer_cursors, and replays the canonical chain.
   * Marks rows finalized = true once current_block - event_block >= confirmation_depth (e.g., 12 blocks on Base L2).
2. **L1 Merkle Batch Worker (SD-10, H-05):**
   * Every 24 hours or whenever COUNT(*) >= 1000 finalized provenance_events have merkle_batch_id IS NULL:
   * Computes a sorted Merkle tree over each event's leaf_hash = keccak256(abi.encode(chainId, contract, txHash, logIndex, tokenId, eventType, blockNumber)).
   * Checks Ethereum L1 base fee; if L1 gas is below the configured ceiling, submits MerkleAnchor.anchorRoot(root, fromBlock, toBlock) on Ethereum L1 and records l1_tx_hash in merkle_batches.

3.5 Layer 5: ERC-4337 Paymaster & Hybrid Meta-Transaction Relayer (server/paymaster/ -- Phase B5)

Implements the gas sponsorship policy defined in SD-09 and ADR-01:

Sponsorship Tier	Allowed Target & Function Selectors (K-01)	Eligible Roles	Default Quota (K-02)	Fallback Behavior (K-04)
Tier T1 (High Priority Escrow Settlement)	ArthenticEscrow.claim() (C-06) ArthenticEscrow.cancelListing() (C-03*)	Buyer / Collector, Artist	10 sponsored txs / month / wallet (max_gas_wei = 350,000)	Prompts Self-Pay Base L2 modal (~$0.01)
Tier T2 (Lifecycle & Curation)	ArthenticEscrow.createListing() (C-01, C-10) ArthenticNFT.updateCondition() (E-03) ArthenticNFT.genesisMint() (B-08)	Accredited ARTIST_ROLE, CONSERVATOR_ROLE, VERIFIER_ROLE	25 sponsored txs / month / wallet	Prompts Self-Pay Base L2 modal or Gallery Safe gas
Tier T3 (Public Verification Anchoring)	ArthenticNFT.recordVerification() (D-04)	Any authenticated wallet	5 sponsored txs / month / wallet (Off-chain D-14* is unlimited & free)	Continues with free off-chain verification or Self-Pay on-chain

3.6 Layer 6: Institutional Web Consoles (web/ / Flutter Web -- Phase B6)

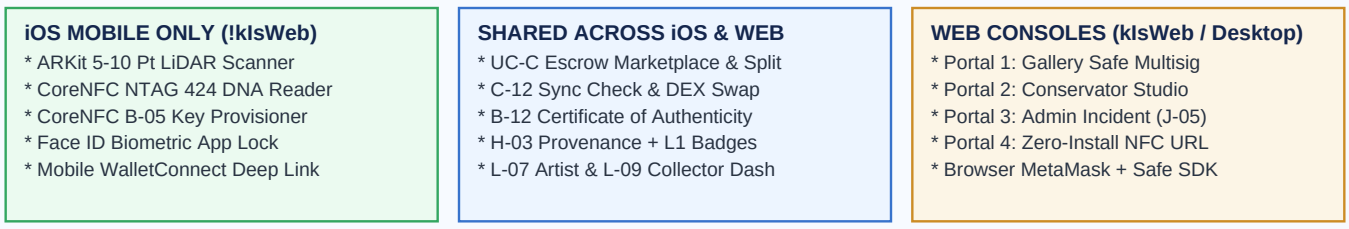
Provides desktop browser interfaces for multi-signature institutional workflows, bulk operations, and protocol security governance (detailed in Part III below).

4. Part III: The Web Client -- Portals, Workflows & Role Matrix

4.1 Why the Web Client is Essential to Arthentic

While the iOS Mobile App is mandatory for physical field interactions requiring iPhone Pro LiDAR (ARKit) and CoreNFC (NTAG 424 DNA), institutional actors operate on desktop workstations for governance, multi-signature execution, high-resolution documentation, and incident monitoring.

ARCHITECTURE DIAGRAM 4: UNIFIED FLUTTER CLIENT -- iOS MOBILE VS. DESKTOP WEB CONSOLES



4.2 Portal 1: Gallery & Verifier Institutional Console (VERIFIER_ROLE + Gnosis Safe)

* **Target Actor:** Galleries, Auction Houses, and Museums holding VERIFIER_ROLE inside a 2-of-3 or 3-of-5 Gnosis Safe Multisig.
* **Key Desktop Screens:**
  1. **Intake Dossier Queue (B-01-B-08):** View scans captured by gallery staff on iPhones in the field, attach high-resolution studio photography and provenance PDFs (B-04), monitor whether the Artist has completed their EIP-712 co-signature (B-07), and propose/sign ArthenticNFT.genesisMint() via Safe{Wallet} + MetaMask Browser Extension (B-08).
  2. **3-of-3 Chip Replacement Coordinator (F-05):** Track Collector (1/3) and Artist (2/3) signatures, attach the Verifier's 3/3 signature, and execute replaceChip() via the Gallery Safe.
  3. Bulk Collection Verifier (/v1/gallery/batch-verify): Audit an entire exhibition or auction catalog's on-chain status (Clear, Flagged, Unsynchronized Ownership, L1 Anchored) and export signed institutional PDF audit reports.

4.3 Portal 2: Conservator Studio Console (CONSERVATOR_ROLE)

* **Target Actor:** Accredited art restorers and conservation laboratories.
* **Key Desktop Screens:**
  1. **Restoration Job Inbox (E-01):** Accept incoming restoration requests from Collectors and inspect historical genesisHash and prior ConditionUpdated records (E-06).
  2. **Treatment Dossier Builder (E-04):** Upload high-resolution before/after microscopy, UV fluorescence photos, and conservation treatment PDFs to Arweave/IPFS (noteURI), paired with the post-restoration 8-point LiDAR vector (newVectorHash) captured via the Conservator's iPhone.
  3. Owner Consent & Condition Update Execution (E-07*, E-03): Request EIP-712 OwnerConsent from the Collector and, once signed, execute ArthenticNFT.updateCondition() via browser MetaMask.

4.4 Portal 3: Protocol Admin Governance & Incident Console (DEFAULT_ADMIN_ROLE)

* **Target Actor:** Arthentic Protocol Governance Council (Safe Multisig + Timelock).
* **Key Desktop Screens:**
  1. Accreditation & Role Governance Queue (A-06, A-07, A-10): Review artist_applications portfolios, gallery KYB credentials, and conservator certifications; propose batch grantRole() or revokeRole() Safe transactions.
  2. Real-Time Security Incident Dashboard (J-05, SD-11): Live stream of security alerts (SUN counter replay bursts, repeated failed claim() attempts, indexer reorgs, or Node signer anomalies) with 1-Click Safe Emergency Actions:
     * **Emergency Pause (J-01):** Proposes pause() on ArthenticNFT and ArthenticEscrow.
     * **Node Signer Key Rotation (A-09, A-10):** Proposes atomic revokeRole(PROTOCOL_NODE_ROLE, compromisedKey) + grantRole(PROTOCOL_NODE_ROLE, newHsmKey).
  3. **Paymaster Treasury & Quota Console (K-01-K-05):** Live EntryPoint deposit gauge (K-03), sponsored_txs burn-rate charts (K-05), and policy editor for Tiers T1-T3.

4.5 Portal 4: Zero-Install Public NFC Tap & CoA Web Viewer (D-14*, B-12)

* **Target Actor:** Museum visitors, art fair attendees, or prospective buyers who tap an NTAG 424 DNA chip with their smartphone without having the Arthentic app installed.
* **How It Works:** Tapping the painting's NFC chip launches https://verify.arthentic.io/?uid=...&ctr=...&cmac=... directly in mobile Safari/Chrome. The Flutter Web / lightweight web route validates the SUN token via POST /v1/verify (D-14*), displays the L1 Hardware Authenticity Result (D-09), renders the Certificate of Authenticity (B-12) and Provenance Timeline (H-03), and offers a deep-link button ("Open in Arthentic iOS App for L2 3D LiDAR Surface Scan").

5. Part IV: Implementing Everything in One Unified Codebase (Monorepo Blueprint)

5.1 Architectural Strategy: 3 Pillars in 1 Repository

Yes -- all client applications (Mobile + Web), all backend/protocol services (BFF + Protocol Node + Indexer + Paymaster), and all smart contracts (Base L2 + Ethereum L1) can and should live in this single repository (Arthenthic-Mobile):

Pillar	Directory	Technology Stack	Unification Mechanism
1. Unified Client (Mobile + Web)	app/	Flutter 3.x (Dart) targeting iOS (!kIsWeb) and Web (kIsWeb)	Single Dart codebase sharing 100% of models, API clients, theme, auth, marketplace, CoA, and role flows. Conditional platform adapters isolate ARKit/CoreNFC on iOS and window.ethereum/Safe SDK on Web.

Pillar	Directory	Technology Stack	Unification Mechanism
2. Unified Backend (Modular Monolith)	server/	Python 3.11+ / FastAPI / SQLAlchemy 2.0 / Redis	Single Python package containing routes/ (BFF), node/ (SUN + ML + KMS signer), paymaster/ (K-01-K-05), and indexer/ (SD-12 + H-05 background workers) sharing one db/models.py schema.
3. Smart Contract Workspace	contracts/	Solidity ^0.8.24 / Foundry (forge)	Lives alongside app/ and server/; exports compiled JSON ABIs directly into server/contracts/abi/ and app/lib/contracts/abi/.

5.2 Pillar 1: Unified Client (app/ -- Flutter iOS + Flutter Web)

1. Platform Capability Abstraction (app/lib/core/platform_capabilities.dart)

By isolating hardware-dependent APIs behind a clean capability service, the rest of the Flutter app compiles seamlessly for both iOS and Web:

CODE / STRUCTURE (DART)

```
import 'package:flutter/foundation.dart' show kIsWeb, defaultTargetPlatform, TargetPlatform;

class PlatformCapabilities {
  static bool get isWeb => kIsWeb;
  static bool get isIOS => !kIsWeb && defaultTargetPlatform == TargetPlatform.iOS;

  /// True only on physical iOS devices with ARKit / Pro LiDAR support (ADR-06)
  static bool get supportsLidarScan => isIOS;

  /// True on iOS devices with CoreNFC NTAG 424 DNA reader/writer support
  static bool get supportsHardwareNfc => isIOS;

  /// True on Desktop/Web layouts (>= 900px width) for Gallery Safe & Admin Consoles
  static bool isDesktopConsole(double screenWidth) => kIsWeb || screenWidth >= 900;
}
```

2. Adaptive Navigation Shell (app/lib/main.dart)

* **Compact Mobile View (width < 900px on iOS):** Renders the existing luxury dark-gallery bottom navigation bar (Home, Scan, Market, Provenance, Wallet, Settings).
* Expanded Web Console View (width >= 900px on Flutter Web): Renders a persistent left NavigationRail / sidebar that includes all shared tabs plus role-gated institutional console tabs:
  * **Gallery Safe Console (VERIFIER_ROLE):** Intake Dossier Manager, Safe Mint Queue (B-08), 3-of-3 Chip Replacement Coordinator (F-05), Batch Verifier.
  * **Conservator Studio (CONSERVATOR_ROLE):** Restoration Requests (E-01), Arweave Dossier Uploader (E-04), Condition Update Signer (E-03).
  * **Protocol Admin Console (DEFAULT_ADMIN_ROLE):** Role Accreditation Queue (A-06/A-07), Live Security Incident Monitor (J-05), Emergency Pause (J-01), Paymaster Ledger (K-03/K-05).

3. Unified Web3 Wallet Adapter (app/lib/services/wallet_adapter.dart)

* **On iOS (!kIsWeb):** Uses WalletConnect v2 (metamask://wc) deep links and iOS Passkey WebAuthn (ASAuthorizationController).
* **On Web (kIsWeb):** Uses injected EIP-1193 provider (window.ethereum for MetaMask Browser Extension), WebAuthn browser passkeys, and @safe-global/safe-apps-sdk JS interop when loaded inside the Gnosis Safe web interface.

5.3 Pillar 2: Unified Backend (server/ -- FastAPI Modular Monolith)

Instead of managing 4 separate repositories and Docker images during development, server/ runs as a Modular Monolith where each architectural layer is a clean Python sub-package that can run in-process (via FastAPI lifespan tasks) or be scaled out as independent worker processes using the exact same codebase:

ARCHITECTURE DIAGRAM 5: UNIFIED PYTHON FASTAPI MODULAR MONOLITH (server/)



* **Single Command for Development:**uvicorn server.main:app --reload starts the BFF API, Protocol Node endpoints, Paymaster endpoints, and background Chain Indexer / L1 Merkle Batcher loops together.
* **Production Process Isolation (Same Codebase):** In production Kubernetes/ECS, the exact same container image can be started with:
  * **uvicorn server.main:** app (API + BFF)
  * python -m server.node.worker (Dedicated HSM-isolated Protocol Node)
  * python -m server.indexer.worker (Dedicated Reorg-Safe Indexer & L1 Merkle Batcher)
  * python -m server.paymaster.worker (Dedicated ERC-4337 Paymaster service)

5.4 Pillar 3: Smart Contract Workspace (contracts/ -- Foundry Solidity)

Adding contracts/ to the root of Arthenthic-Mobile keeps Solidity contracts, Python backend verifiers, and Dart mobile/web clients 100% synchronized on EIP-712 typehashes, struct layouts, and contract ABIs:

* A single script (scripts/sync_abis.py) compiles contracts/src/*.sol via forge build and writes the ABI JSONs to server/contracts/abi/ and app/lib/contracts/abi/.

5.5 Complete Unified Monorepo Directory Structure

CODE / STRUCTURE (TEXT)

```
Arthenthic-Mobile/
|+-- app/                                  # PILLAR 1: UNIFIED FLUTTER CLIENT (iOS + Web)
|   |+-- ios/                              # iOS native runner (ARKit LiDAR + CoreNFC channels)
|   |+-- web/                              # Flutter Web runner (MetaMask EIP-1193 + Safe SDK)
|   '--- lib/
|       |+-- main.dart                     # Adaptive shell (BottomNav on iOS, Sidebar on Web)
|       |+-- core/
|       |   |+-- platform_capabilities.dart    # kIsWeb vs iOS Pro LiDAR / CoreNFC guards
|       |   |+-- theme.dart                    # Museum dark-gallery & brass plaque design system
|       |   '--- api_client.dart               # Shared Dio/HTTP client + JWT refresh interceptor
|       |+-- contracts/
|       |   '--- abi/                          # Auto-synced ABIs (ArthenticNFT, ArthenticEscrow)
|       |+-- services/
|       |   |+-- wallet_service.dart           # Unified MetaMask (WCv2 / EIP-1193) + Passkey + Safe
|       |   |+-- nfc_service.dart              # NTAG 424 DNA SUN parser & B-05 key writer
|       |   '--- app_lock_service.dart         # Face ID / passcode guard (mobile)
|       '--- features/
|           |+-- auth/                         # Login, Register, OAuth, Biometric Lock
|           |+-- genesis/                      # B-01 Intake, B-07 Artist Co-Sign, B-12 Public CoA
|           |+-- scan/                         # 5-10 Pt Guided LiDAR Scan (iOS) / Upload (Web)
|           |+-- result/                       # D-09 Verification Result + D-15* Status Badges
|           |+-- marketplace/                  # UC-C Escrow Market, C-12 Check, ETH/USDC DEX Swap
|           |+-- conservation/                 # E-01 Request, E-07* Owner Consent, F-05 3-of-3 Swap
|           |+-- provenance/                   # H-03 Timeline + H-05 Ethereum L1 Merkle Proof Badge
|           |+-- dashboards/                   # L-07 Artist Royalty & L-09 Collector Dashboards
|           |+-- wallet/                       # Multi-wallet manager & K-04 Self-Pay Gas Modal
|           '--- web_consoles/                 # Desktop/Web-optimized institutional views:
|               |+-- gallery_safe_console.dart # B-08 Safe Mint, F-05 Replace Chip, Batch Verify
|               |+-- conservator_console.dart  # E-04 Arweave Dossier Builder, E-03 Condition Update
```

CODE / STRUCTURE (TEXT) [CONT.]

```
|              '--- admin_incident_console.dart # J-05 Incident Stream, J-01 Pause, A-06..10 Roles
|
|+-- server/                                   # PILLAR 2: UNIFIED PYTHON MODULAR MONOLITH
|   |+-- main.py                               # FastAPI entry + lifespan background workers
|   |+-- db/
|   |   |+-- models.py                         # 6-Domain SQLAlchemy ORM (PostgreSQL 16 + SQLite dev)
|   |   '--- redis_hotpath.py                  # 6 Redis atomic keys (sun:ctr, nonce, quota, etc.)
|   |+-- crypto/
|   |   |+-- sun.py                            # AES-128 CMAC & monotonic counter verifier
|   |   '--- attestation.py                    # EIP-712 typed data hashing & signature recovery
|   |+-- ml/
|   |   '--- embedder.py                       # 384-D weave + crack + LiDAR depth similarity engine
|   |+-- node/                                 # Phase B3: Protocol Node Layer
|   |   |+-- hsm_signer.py                     # Cloud KMS / Vault Transit EIP-712 & AES-128 adapter
|   |   '--- arweave_storage.py                # Permanent Arweave/IPFS uploader & hash verifier
|   |+-- indexer/                              # Phase B4: Chain Indexer & L1 Batcher Layer
|   |   |+-- event_listener.py                 # Reorg-safe Base L2 event subscriber (SD-12)
|   |   |+-- merkle_batcher.py                 # Ethereum L1 Merkle root batch worker (H-05)
|   |   '--- reconciler.py                     # Periodic on-chain ownerOf/genesisHash drift healer
|   |+-- paymaster/                            # Phase B5: ERC-4337 Paymaster Layer
|   |   |+-- policy_engine.py                  # K-01 allow-list (T1-T3) & K-02 per-user quotas
|   |   '--- balance_monitor.py                # K-03 EntryPoint deposit monitor & K-05 cost ledger
|   |+-- routes/                               # BFF REST API Endpoints
|   |   |+-- auth.py                           # Hybrid Auth, OAuth, Sessions & MetaMask SIWE
|   |   |+-- genesis.py                        # B-01 Intake, B-07 Co-Sign, B-08 Mint, B-12 CoA
|   |   |+-- marketplace.py                    # UC-C Listings, Escrow, 93.5/5/1.5 Split & DEX Swap
|   |   |+-- conservation.py                   # UC-E Restoration, E-07* Consent & UC-F 3-of-3 Swap
|   |   |+-- verify.py                         # UC-D L1/L2 Verification & D-15* Status Badges
|   |   |+-- roles.py                          # Role Applications & On-Chain Grant Sync
|   |   '--- admin.py                          # J-05 Incidents, J-01 Emergency Pause & K-05 Metrics
|   '--- tests/                                # Comprehensive Pytest Suite
|
|+-- contracts/                                # PILLAR 3: FOUNDRY SOLIDITY SMART CONTRACTS
|   |+-- foundry.toml                          # Base L2 + Ethereum L1 RPC & compiler config
|   |+-- src/
|   |   |+-- Roles.sol                         # Role constants library
|   |   |+-- AttestationVerifier.sol           # EIP-712 verifier, 8800 bps gate, usedNonce map
|   |   |+-- ArthenticNFT.sol                  # UUPS ERC-721 + ERC-2981 + ADR-02/03 enforcement
|   |   |+-- ArthenticEscrow.sol               # Escrow state machine + ADR-04/05 93.5/5/1.5 split
|   |   '--- MerkleAnchor.sol                  # Ethereum L1 batch Merkle root anchor
|   |+-- test/                                 # Foundry unit, fuzz, and invariant tests
|   '--- script/                               # Deploy scripts for Base Sepolia & Eth Sepolia
|
'--- docs/                                     # Master Architecture & Security Specifications
```